Medhansh Garg

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Education

University of Illinois, Urbana-Champaign Aug. 2023 – May 2027 (Expected)
- B.S. in Computer Engineering | Coursework: Data Structures & Algorithms, Computer Systems Engineering, Computer Architecture, Operating Systems, Distributed Systems, Computer Networks, Databases

Skills

- **Languages**: Python, TypeScript/JavaScript, C, C++, Java, Kotlin, Go, Dart, SQL
- **Web & Backend**: React, Next.js, Node.js, FastAPI, Spring Boot, Prisma, PostgreSQL, Supabase, Flutter, REST APIs
- **Cloud & DevOps**: Linux, Docker, Kubernetes, AWS, Azure, GitHub Actions CI/CD, Vercel, nginx, Tailscale, Git
- **ML & AI**: PyTorch, MediaPipe, ONNX, OpenCV, YOLOv8, LLM agents (Claude Code), Vitest, Playwright, pytest

Experience

Hopcharge, Gurugram, India | *Software Engineering Intern* Jun. 2026 – Aug. 2026
- Architected the Ad Engine, an end-to-end ad pipeline: AI-generated creatives, human review, Meta and YouTube publishing, and a performance dashboard feeding winners back into new ads; scaled output **8x** (2–3/week to **3/day**)
- Developed VOLT.CV to automate hiring end to end: resume parsing, Gmail outreach, reply classification, bulk interview scheduling; secured candidate data with row-level security on Google identities; shipped as a one-click desktop app for **4 OS/architectures**
- Consolidated **5 manual scripts** into Grid Ops, a FastAPI/React mapping service forecasting battery-swap demand and recommending charging-pod placement across **3 Delhi-NCR metros**

CS 124 Honors, Urbana, IL | *Lead Web Developer* Aug. 2025 – Present
- Lead an **11-person** team building the course platform for **300+ students** and **30+ staff**; run weekly planning and PR reviews
- Improved Lighthouse score from **76 to 99**; implemented CILogon SSO with 6 least-privilege roles; moved all database access server-side, keeping credentials out of the browser
- Designed a **246-test** suite (unit, API/DB integration, Playwright E2E) gating every release in CI; automated grading, roster imports and attendance via 30-second rotating codes

Care Health Insurance, Gurugram, India | *Cybersecurity Intern* Jun. 2025 – Aug. 2025
- Discovered **60+ unauthenticated internet-exposed services**, **30** pulled offline after my report; reverse-engineered WAF request signing into a working replay exploit; bypassed SSL pinning and root detection for live testing of the customer Android app

IoT++, Remote | *Software Engineering Intern* Jun. 2024 – Aug. 2024
- Developed real-time fire/smoke detection, improving accuracy **62%** over base YOLOv8 and inference speed **13x** via pruning and tiling; deployed to **12 edge devices** in a factory pilot (Docker, Kubernetes, Azure)
- Trained Random Forest and RNN models routing trucks to weighbridges from live traffic and equipment throughput

Projects

ASL Adaptive Tutor | *PyTorch, MediaPipe, Flutter, ONNX* github.com/HackOverflow404/SignLanguageLearning
- Engineered a computer-vision ASL tutor that pinpoints which part of a sign is wrong (handshape, location, movement) and adapts practice; trained grader hits **85%** accuracy across **60 signs**, **2.7x** the baseline
- Ported to iPhone with on-device grading validated against the Python pipeline (**400+ tests**); added LLM-generated sentence practice with a second-model filter, **96%** sensible output

Pensieve: AI Note Librarian | *Python, Claude Code* hackoverflow404.github.io/Portfolio/experience?entry=pensieve
- Built a self-hosted knowledge base with one-tap phone capture and an AI agent that files, links and organizes notes every 15 min; **125 notes** processed over **68 runs** in its first 2 weeks
- Engineered for unattended operation: sandboxed agent, rule-based checker rejecting any run that alters original text, and each run a single revertible git commit the agent learns from

Clippy: Jailbroken Echo Show | *C++, Kotlin, Go, Android* github.com/HackOverflow404/EchoCameraStreamer
- Replaced Amazon's OS with a source-built LineageOS ROM; diagnosed a dead camera as a vendor-library ABI mismatch; built a self-healing webcam (**55s** reboot recovery, idle CPU **12% to 0.1%**)
- Developed a private voice assistant with a custom-trained wake word and on-device speech models; patched Android's Bluetooth stack, stabilizing speaker latency at **30–180ms**
- Converted its camera into a security camera recording **18h** of rolling footage to a NAS at ∼**7%** Pi CPU

TPS-L2: Spotify Wall Display | *JavaScript, Vercel, Redis, Raspberry Pi* github.com/HackOverflow404/TPS-L2
- Repurposed a salvaged laptop screen as a now-playing display with synced lyrics; designed QR sign-in for keyboardless screens, cutting database usage ∼**10x** with signed, self-expiring codes

ECE 391 RISC-V Kernel | *C, RISC-V, WebAssembly* ece-391-kernel-demo.vercel.app
- Co-developed an OS kernel from scratch (drivers, filesystem, virtual memory, scheduler, syscalls, shell with pipes); compiled an emulator to WebAssembly so it boots live in any browser

Leadership & Involvement

Illinois MicroTech, Urbana, IL | *External Vice President* | *EOH Project Lead* | *HackerFab Automation Member* Oct. 2025 – Present
- Run external communications, overseeing the outreach team, marketing chairs, website and EOH chairs
- Led a **15-person** team building a 4-stage interactive MEMS exhibit for **20,000+** Engineering Open House visitors on a **$250** budget
- HackerFab: automating the wafer stage of a student-built chip-fabrication rig for autonomous full-wafer movement